FSU | DEPARTMENT OF COMPUTER SCIENCE
COLLEGE OF ARTS & SCIENCES

CIS 4930/5930 Future Edge Networks and Distributed Intelligence

Fall 2026

Course Information

Course:	CIS 4930/5930 Future Edge Networks and Distributed Intelligence
Credit Hours:	3
Lecture Room:	LOV 0307
Lecture Day/Times:	Monday, Wednesday, 1:20 p.m. - 2:35 p.m.
Course Website:	Course website

Contact Information

Instructor:	Dr. Xin Liu
Email:	xliu15@fsu.edu
Office Hours:	Monday, 3:30 p.m. - 5:30 p.m. Available via Zoom (appointment required).
Office:	James Jay Love Building, 160
Teaching Assistant:	Zhankai Ye zy22b@fsu.edu

Course Objectives

- **Fundamental Goal: Applying AI/ML to 5G and Beyond**
  - **AI/ML for Wireless Networks:** Understand how modern machine learning, especially deep learning, can be used to improve the design and operation of 5G and beyond wireless systems.
  - **System-Level Thinking:** Connect networking principles with ML pipeline design, focusing on how learning components interact with real network constraints.
- **Core Focus: 5G/ORAN and Hybrid Routing Systems**
  - **5G Architecture and ORAN:** Learn the overall 5G network architecture with an emphasis on the Open Radio Access Network (ORAN), its key interfaces, and deployment considerations.
  - **Model Routing in Edge Environments:** Design mechanisms to select, place, and invoke ML models dynamically across the network, balancing latency, accuracy, and resource constraints.
  - **Hybrid Routing Design:** Study how model routing and network routing interact, and learn to build integrated systems that jointly optimize both.
- **Key Topics and Emphasis on Edge-Side Challenges**
  - **Edge Resource Constraints:** Analyze compute, memory, energy, and bandwidth limitations at the edge, and their implications for real-time ML inference and control.
  - **Wireless Fundamentals for Edge Intelligence:** Cover essential concepts such as OFDM and resource allocation, focusing on how they shape edge-side ML performance.
  - **Selective Coverage of the 5G Core:** Briefly introduce core-network concepts (e.g., slicing strategies) to provide context, while keeping the primary emphasis on edge-side challenges.

Course Prerequisites

While no prior experience with wireless systems is required, students should be comfortable with programming (**MatLab** and **Python** will be used frequently) and have a basic understanding of machine learning, including some hands-on experience with deep learning techniques.

Course Material

- **Textbook:** Editors: Afif Osseiran, José F. Monserrat, Patrick Marsch, 5G Mobile and Wireless Communications Technology.
  (Note: Free and legal electronic versions are available online.).
- **Slides and Codes:** Available directly on the course website.

Computer Accounts

- **GitHub Account:** You will need a GitHub account to access course assignments and use GitHub Codespaces.
- **GitHub Student Developer Pack:** It is strongly recommended to apply for the GitHub Student Developer Pack to access free credits for Codespaces and other developer tools.
- **FSU Email:** You will need a `my.fsu.edu` account for receiving class emails. For more information, visit the FSU email account management page.

Your Responsibilities

- Check the course website and your email account regularly
- Understand the lecture slides and reading assignments
- Uphold academic honesty in completing your assignments, projects, and exams
- Turn in your projects on time
- Attend office hours for extra help, as needed

Course Calendar

Class	Date	Topic	Notes
Introduction			
1	8/24 (M)	Course Syllabus Preparation for Learning Edge Networks (XG History and Course Tools)	Mandatory first-day attendance. **No-shows will be dropped.**
Don't rush to test the speed; first, let's understand the 'G' in 5G.			
2	8/26 (W)	XG Physical Layer (CDMA, OFDM, MIMO, Beamforming, mmWave) Recommended reading: MATLAB 5G Toolbox	HW 1 posted on Canvas
3-4	8/31 (M) 9/2 (W)	XG Architecture Basics (Generations, Mobility, RAN/Core, O-RAN, Security, Market) Recommended reading: How to deploy "antivirus software" on each 5G base station?	9/2 (W): HW 1 DDL at 11:59 p.m. (ET) 9/2 (W): HW 2 posted on GitHub Classroom
-	9/7 (M)	No class due to Labor Day.	

Class	Date	Topic	Notes
5	9/9 (W)	Integrated Sensing and Communication (ISAC) (What it is, why it is hard, and how OFDM does it) Recommended reading: ISAC Report	
6	9/14 (M)	Paper Practicum 1 (XG I): Inside the Base Station — Software and Silicon (Two student presentations, 15 min each + 5 min Q&A) Papers: SwapRAN; MegaStation	
7	9/16 (W)	Paper Practicum 2 (XG II): Operating the Base Station — Efficiency, AI, Adversary (Three student presentations, 15 min each + 5 min Q&A) Papers: RENC; SAGE; SNI5GECT	HW 2 DDL at 11:59 p.m. (ET)
You've used Wi-Fi your whole life. It's far more fun than you think — and after today, you'll know it better than anyone.			
8-9	9/21 (M) 9/23 (W)	Wi-Fi History (802.11b to 802.11be, PHY Waveforms, Spectrum, MIMO, OFDMA, the 60 GHz ad/ay Branch) Recommended reading: IEEE Wi-Fi Timeline	9/23 (W): HW 3 posted on GitHub Classroom
10	9/28 (M)	Paper Practicum 3 (Wi-Fi I): Why Your Video Call Stutters on Wi-Fi (Two student presentations, 15 min each + 5 min Q&A) Papers: Zhuge; AFR	
11	9/30 (W)	Paper Practicum 4 (Wi-Fi II): Inside 802.11 — Which Knobs Can We Still Redesign? (Three student presentations, 15 min each + 5 min Q&A) Papers: Law; BLADE; SMUFF	
Communication Was Only the Radio's First Job. The Same Wave That Carries Your Bits Can Also See the Room.			
12	10/5 (M)	Wi-Fi Sensing (Where Is the Phone? Signal Strength, Phase, Angle with ArrayTrack, Distance with Chronos, Both at Once with SpotFi) Recommended reading: Hands-on Wireless Sensing: CSI Feature Extraction (Tsinghua)	
13	10/7 (W)	Paper Practicum 5 (ISAC): From Reference Signals to Vital Signs (Three student presentations, 15 min each + 5 min Q&A) Papers: CARTS; CommRad; 2FiA	HW 3 DDL at 11:59 p.m. (ET)
14	10/12 (M)	Using 802.11 Beamforming Feedback as a Sensing Signal (Why Beamform, How the AP Asks: NDPA and NDP, From CSI to the Compressed BFM, Angles with MUSIC, the Same Angles from the BFM) Live demo: tracking people with a Pixel Watch 3 — BFM-Scan (ACM UbiComp 2026)	HW 4 posted on GitHub Classroom
Don't Be Afraid of Artificial Intelligence. It's Not That Intelligent.			
15	10/14 (W)	Core Models of Modern AI (CNN, RNN, Transformer, LLM, Agent, Diffusion) Recommended reading: Dive into Deep Learning	
16	10/19 (M)	PyTorch Tutorial: From Colab to a Trained Model (Google Colab with a GPU, Tensors, the Training Pipeline: Data, Network, Loss, Optimizer, Train/Validate/Test; Live Demo: Choosing a Neural Network, CNN vs. RNN vs. Transformer) Recommended reading: PyTorch: Learn the Basics	

Class	Date	Topic	Notes
17	10/21 (W)	Semantic Communication (Beyond bits: task-oriented and knowledge-driven communication — why transmit every bit when the receiver only needs the meaning?) (One student presentation, 15 min + 5 min Q&A) Papers: CoOrbit	
18	10/26 (M)	Paper Practicum 6 (Semantic Communication I): The Representation Shrinks (Three student presentations, 15 min each + 5 min Q&A) Papers: Empilo; MagicStream; Gemino	
19	10/28 (W)	Paper Practicum 7 (Semantic Communication II): From Pixels to Tokens, and Surviving Loss (Three student presentations, 15 min each + 5 min Q&A) Papers: KDC; NIER; Artic	
The Cloud Is Far Away. The Phone Is Small. Edge Intelligence Is Deciding What Runs Where.			
20	11/2 (M)	Edge AI Systems: Where Should the Model Run? (Why On-Device AI, Memory/Compute/Energy Limits, Quantization and Mixture-of-Experts, KV Cache, Split and Collaborative Inference, Model Routing) (One student presentation, 15 min + 5 min Q&A) Papers: MobileMoE	
21	11/4 (W)	Paper Practicum 8 (Edge AI I): Making AI Fit on the Device (Three student presentations, 15 min each + 5 min Q&A) Papers: ElastiLM; D2MoE; KVSwap	HW 4 DDL at 11:59 p.m. (ET) Final Project Release
22	11/9 (M)	Paper Practicum 9 (Edge AI II): Where Should Intelligence Run? (Three student presentations, 15 min each + 5 min Q&A) Papers: SAIL; SMEC; Confidant	
-	11/11 (W)	No class due to Veterans Day observed.	
Seeing the Room Is Not Enough. An Edge Agent Has to Act — on Time, Safely, and Without Giving You Away.			
23	11/16 (M)	Edge AI Agents: From Perception to Action (The Agent Loop, End-to-End Latency Budgets, Choosing What to Sense, Checking Actions Before They Run, Privacy)	
24	11/18 (W)	Paper Practicum 10 (Edge Agents I): From Perception to Action (Three student presentations, 15 min each + 5 min Q&A) Papers: Modality Plug-and-Play; TimelyLLM; VeriSafe Agent	
25	11/23 (M)	Paper Practicum 11 (Edge Privacy): What the Signal, the Features, and the Cache Give Away (Three student presentations, 15 min each + 5 min Q&A) Papers: Poison to Cure; FaceObfuscator; KV-Cloak	
-	11/25 (W)	No class due to Thanksgiving holiday.	
26	11/30 (M)	Paper Practicum 12 (Edge Agents II): Agents That See, Sense, Tap, and Audit (Four student presentations, 15 min each + 5 min Q&A) Papers: SpatialMind; IoT-Brain; V-Droid; 5GAudit	
-	12/2 (W)	No class. Use the time to work on your final project.	
Final Exam Week			

Class	Date	Topic	Notes
-	12/7 (M) 12/9 (W)	No class, no final exam.	12/9 (W): Final Project DDL at 11:59 p.m. (ET)

The course calendar is tentative. Please refer to the course website for the most up-to-date lecture topics, assignment deadlines, and course notices.

Course Evaluation and Grading Policies

- **Assignments: 45%**

There will be several homework assignments (written and coding-based) spaced out over the course of the semester. All assignments are equally weighted. Submission instructions will be posted on GitHub Classroom and/or Canvas; most project submissions use GitHub.

- **Paper Practicum (Reading & Presentation): 15%**

Each student will complete **one** in-class paper practicum on top-tier conference papers related to **Future Edge Networks**.

(1) **Format: Individual presentation only** (one person per practicum). The same time allocation applies to both undergraduate and graduate students: **15 minutes** presentation + **5 minutes** Q&A.

(2) **Scheduling: Three** papers are presented per class meeting. The practicum block runs for roughly seven weeks after the introductory lectures conclude.

(3) **Paper Assignment:** Papers will be assigned by the instructor. Students will first submit their preferred presentation dates (and optionally topics of interest). The instructor will then assign a paper to each student based on the course schedule and topic sequence.

Important policy for paper practicum:

(1) **In-person presentation only.** No make-up or deferred practicums will be approved.

(2) Presentation time slots will be assigned in random order. Plan ahead when submitting your availability.

- **Final Project: 40%**

Build an AI-based sensing system on the Wi-Fi beamforming feedback (BFM) you capture yourself, continuing from HW 4. Everything is due by **11:59 p.m. (ET) on Wednesday, 12/9/2026**. There is no final presentation.

(1) **Code and data (20%)**: push them to your final-project repository on GitHub.

(2) **Demo video (10%)**: a 3–5 minute video of your system working; you may talk over your slides. Upload it to Canvas.

(3) **Paper (10%)**: up to 6 pages in the format of ACM HotMobile, references not counted, written in the official two-column ACM template (Overleaf Template). Posting it on arXiv is required: submit it to arXiv yourself, then paste its arXiv link or submission ID on Canvas. If arXiv asks you for an endorsement, ask the instructor.

Please see a detailed introduction of the final project on the course site.

Extra Bonus.

Students are encouraged to prepare submissions to major AI/ML/DM conferences based on their projects. Please make an appointment with the instructor prior to any submission plan for a comprehensive evaluation of the research topic. Each submission under the instructor's recognition will gain **2 points** on the final grade.

Letter grades are assigned based on the final percent using the cut-off values:

Grade	**% bound**
A	93 - 100
A-	90 - 92.9
B+	87 - 89.9
B	83 - 86.9
B-	80 - 82.9
C+	77 - 79.9
C	73 - 76.9
C-	70 - 72.9
D	60 - 69.9
F	<60

Course Policies, Support Services, and Other Information

Missed Exam Policy

Unexcused missed exams and homework will be given a grade of 0. See the University Attendance Policy for a discussion of valid reasons to excuse absences: `https://registrar.fsu.edu/bulletin/graduate/information/academic_regulations/`.

Grade of "I" Policy

Incomplete ("I") grades should be recorded only in exceptional cases when a student, who has completed a substantial portion of the course and who is otherwise passing, is unable to complete a well-defined portion of a course for reasons beyond the student's control. Students in these circumstances must petition the instructor and should be prepared to present documentation that substantiates their case.

University Attendance Policy

Excused absences include documented illness, deaths in the family and other documented crises, call to active military duty or jury duty, religious holy days, and official University activities. These absences will be accommodated in a way that does not arbitrarily penalize students who have a valid excuse. Consideration will also be given to students whose dependent children experience serious illness.

Academic Honor Policy

The Florida State University Academic Honor Policy outlines the University's expectations for the integrity of students' academic work, the procedures for resolving alleged violations of those expectations, and the rights and responsibilities of students and faculty members throughout the process. Students are responsible for reading the Academic Honor Policy and for living up to their pledge to "...be honest and truthful and...[to] strive for personal and institutional integrity at Florida State University." (`http://fda.fsu.edu/Academics/Academic-Honor-Policy`).

For this course, in particular, every student must complete his/her assignments, quizzes, and exams independently. Showing your work to your peers or making it accessible to them is considered academic dishonesty. You are responsible for ensuring that your work is adequately protected and not accessible to others.

Americans with Disabilities Act
Students with disabilities needing academic accommodation should:

1. Register with and provide documentation to the Student Disability Resource Center (SDRC).
2. Bring a letter to the instructor indicating the need for accommodation and the specific type required.

Please note that instructors cannot provide classroom accommodations until appropriate verification from the SDRC has been received. This syllabus and other class materials are available in alternative formats upon request.
For more information about services available to FSU students with disabilities, please contact the SDRC:

- Address: 874 Traditions Way, 108 Student Services Building, Florida State University, Tallahassee, FL 32306-4167
- Phone: (850) 644-9566 (voice), (850) 644-8504 (TDD)
- Email: sdrc@admin.fsu.edu
- Website: `http://www.disabilitycenter.fsu.edu`

Confidential Campus Resources
Various centers and programs are available to assist students with navigating stressors that might impact academic success. These include the following:

- **Victim Advocate Program:** University Center A, Room 4100, (850) 644-7161, Available 24/7/365, Office Hours: M-F 8-5, `https://dsst.fsu.edu/vap`
- **University Counseling Center:** Askew Student Life Center, 2nd Floor, 942 Learning Way, (850) 644-8255, `https://counseling.fsu.edu/`
- **University Health Services:** Health and Wellness Center, (850) 644-6230, `https://uhs.fsu.edu/`

Free Tutoring from FSU
On-campus tutoring and writing assistance are available for many courses at Florida State University. For more information, visit the Academic Center for Excellence (ACE) Tutoring Services' comprehensive list of on-campus tutoring options at `http://ace.fsu.edu/tutoring` or contact tutor@fsu.edu. High-quality tutoring is available by appointment and on a walk-in basis. These services are offered by tutors trained to encourage the highest level of individual academic success while upholding personal academic integrity.

Late Policy and Make-up Exams

- Late assignments will not ordinarily be accepted. If, for some compelling reason, you cannot hand in an assignment on time, please contact the instructor as far in advance as possible.
- No credit will be given to late course projects.
- No make-up exams (except under extremely unusual circumstances).

Syllabus Change Policy
Except for changes that substantially affect the implementation of the evaluation (grading) statement, this syllabus is a guide for the course and is subject to change with advance notice.